Online Resource 1

Characterizing Scanpath Decoding Pipelines with Perturbation Profiles

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Tables S1–S13: every profile of the primary analysis (within-fold AUC, class-balanced participant folds); $R = 6$ replicates, $B = 60$ refit-bootstrap draws, seed 42; linear SVM, ten-fold participant-held-out cross-validation; duration-marginal donor pool of the other participants' durations. u is the reference scale (one text line in SB-SAT and EMIP, 2° in GazeBase). The same profiles with the pooled AUC are in the released outputs.

Table S1: Profile of `gb_tex_vd1`: reading vs. video viewing. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. $0.25u$	jit. $0.5u$	jit. $1u$	jit. $2u$
Count	0.969	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	1.000	0 (ex.)	0 (ex.)	0 (ex.)	**+0.021 [+0.01, +0.03]**	+0.000 [−0.00, +0.00]	−0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]
Classic	0.969	**−0.022 [−0.03, −0.01]**	0 (ex.)	**+0.035 [+0.02, +0.05]**	0 (ex.)	**−0.003 [−0.01, −0.00]**	**−0.006 [−0.01, −0.00]**	**−0.007 [−0.01, −0.00]**	−0.004 [−0.01, +0.00]
Dur. trend	0.957	+0.003 [−0.00, +0.01]	+0.003 [−0.00, +0.01]	**+0.473 [+0.44, +0.50]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA $0.5u$	0.999	**+0.010 [+0.00, +0.02]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.001 [−0.00, +0.00]	+0.002 [−0.00, +0.00]	**+0.004 [+0.00, +0.01]**	**+0.015 [+0.01, +0.02]**
RQA $1u$	1.000	**+0.006 [+0.00, +0.01]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.000 [−0.00, +0.00]	+0.001 [−0.00, +0.00]	+0.001 [−0.00, +0.00]	**+0.006 [+0.00, +0.01]**
Directed	0.977	**−0.014 [−0.02, −0.01]**	0 (ex.)	0 (ex.)	**−0.007 [−0.01, −0.00]**	**−0.011 [−0.02, −0.00]**	**−0.015 [−0.02, −0.01]**	**−0.016 [−0.02, −0.01]**	**−0.013 [−0.02, −0.01]**
FAG lattice	0.988	0 (ex.)	−0.001 [−0.00, +0.00]	+0.004 [−0.00, +0.01]	+0.002 [−0.00, +0.01]	−0.000 [−0.01, +0.00]	+0.001 [−0.00, +0.01]	**+0.009 [+0.00, +0.02]**	**+0.049 [+0.04, +0.06]**

Table S2: Profile of `gb_vd1_blg`: video viewing vs. ball-catching game. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. $0.25u$	jit. $0.5u$	jit. $1u$	jit. $2u$
Count	0.690	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	1.000	0 (ex.)	0 (ex.)	0 (ex.)	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	−0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]
Classic	0.886	**−0.114 [−0.14, −0.09]**	0 (ex.)	**+0.033 [+0.02, +0.05]**	0 (ex.)	**−0.003 [−0.00, −0.00]**	**−0.007 [−0.01, −0.00]**	**−0.014 [−0.02, −0.01]**	**−0.014 [−0.02, −0.01]**
Dur. trend	0.752	+0.011 [−0.00, +0.02]	+0.011 [−0.00, +0.02]	**+0.252 [+0.20, +0.30]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA $0.5u$	0.999	**+0.013 [+0.01, +0.02]**	0 (ex.)	0 (ex.)	0 (ex.)	−0.000 [−0.00, +0.00]	−0.000 [−0.00, +0.00]	−0.000 [−0.00, +0.00]	**+0.003 [+0.00, +0.01]**
RQA $1u$	1.000	**+0.001 [+0.00, +0.00]**	0 (ex.)	0 (ex.)	0 (ex.)	−0.000 [−0.00, +0.00]	−0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]
Directed	1.000	**+0.005 [+0.00, +0.01]**	0 (ex.)	0 (ex.)	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	+0.000 [−0.00, +0.00]	**+0.002 [+0.00, +0.00]**
FAG lattice	0.957	0 (ex.)	+0.000 [−0.00, +0.00]	**−0.008 [−0.01, −0.00]**	**+0.029 [+0.02, +0.04]**	+0.007 [−0.00, +0.02]	+0.010 [−0.00, +0.02]	**+0.019 [+0.00, +0.04]**	**+0.027 [+0.01, +0.04]**

Table S3: Profile of `gb_vd1_vd2`: video segment 1 vs. 2. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.634	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.707	0 (ex.)	0 (ex.)	0 (ex.)	**+0.098 [+0.07, +0.13]**	+0.004 [−0.00, +0.01]	**+0.012 [+0.00, +0.02]**	**+0.045 [+0.03, +0.06]**	**+0.116 [+0.08, +0.15]**
Classic	0.640	+0.007 [−0.00, +0.01]	0 (ex.)	**+0.124 [+0.10, +0.15]**	0 (ex.)	+0.001 [−0.00, +0.00]	**+0.004 [+0.00, +0.01]**	+0.005 [−0.00, +0.01]	+0.005 [−0.00, +0.01]
Dur. trend	0.790	**+0.157 [+0.13, +0.19]**	**+0.158 [+0.13, +0.19]**	**+0.286 [+0.23, +0.34]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.677	**+0.035 [+0.01, +0.06]**	0 (ex.)	0 (ex.)	0 (ex.)	−0.006 [−0.03, +0.01]	+0.002 [−0.03, +0.03]	**+0.046 [+0.00, +0.09]**	**+0.108 [+0.07, +0.14]**
RQA 1u	0.719	**+0.073 [+0.04, +0.10]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.007 [−0.01, +0.02]	+0.013 [−0.01, +0.03]	**+0.043 [+0.01, +0.08]**	**+0.118 [+0.08, +0.16]**
Directed	0.700	**+0.040 [+0.02, +0.06]**	0 (ex.)	0 (ex.)	−0.010 [−0.03, +0.01]	−0.005 [−0.03, +0.02]	+0.013 [−0.01, +0.04]	**+0.042 [+0.01, +0.08]**	**+0.069 [+0.04, +0.10]**
FAG lattice	0.664	0 (ex.)	**−0.015 [−0.03, −0.00]**	**−0.020 [−0.03, −0.01]**	−0.012 [−0.05, +0.02]	−0.015 [−0.05, +0.02]	−0.006 [−0.05, +0.04]	+0.017 [−0.03, +0.06]	**+0.055 [+0.01, +0.10]**

Table S4: Profile of `gb_vd1_halves`: video 1, first vs. second half. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.500	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.924	0 (ex.)	0 (ex.)	0 (ex.)	**+0.249 [+0.22, +0.27]**	+0.001 [−0.00, +0.00]	**+0.007 [+0.00, +0.01]**	**+0.029 [+0.02, +0.04]**	**+0.130 [+0.11, +0.15]**
Classic	0.564	**−0.090 [−0.11, −0.07]**	0 (ex.)	+0.008 [−0.01, +0.03]	0 (ex.)	−0.001 [−0.00, +0.00]	+0.002 [−0.00, +0.01]	+0.005 [−0.01, +0.02]	+0.015 [−0.00, +0.03]
Dur. trend	0.590	**+0.076 [+0.03, +0.12]**	**+0.085 [+0.04, +0.13]**	**+0.082 [+0.04, +0.12]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.743	**+0.046 [+0.02, +0.07]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.012 [−0.01, +0.03]	+0.025 [−0.00, +0.05]	**+0.079 [+0.05, +0.11]**	**+0.143 [+0.10, +0.19]**
RQA 1u	0.823	**+0.088 [+0.06, +0.11]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.003 [−0.01, +0.02]	**+0.026 [+0.01, +0.05]**	**+0.088 [+0.05, +0.12]**	**+0.182 [+0.15, +0.21]**
Directed	0.805	**+0.036 [+0.02, +0.05]**	0 (ex.)	0 (ex.)	−0.007 [−0.03, +0.01]	+0.010 [−0.01, +0.03]	**+0.033 [+0.01, +0.05]**	**+0.090 [+0.06, +0.12]**	**+0.191 [+0.16, +0.22]**
FAG lattice	0.741	0 (ex.)	−0.003 [−0.01, +0.00]	+0.000 [−0.00, +0.01]	+0.023 [−0.01, +0.05]	+0.004 [−0.02, +0.03]	+0.011 [−0.02, +0.05]	+0.037 [−0.00, +0.07]	**+0.088 [+0.05, +0.12]**

Table S5: Profile of `gb_tex_s1s2`: reading, session 1 vs. 2. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.530	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.519	0 (ex.)	0 (ex.)	0 (ex.)	+0.036 [−0.05, +0.12]	+0.003 [−0.01, +0.01]	−0.000 [−0.03, +0.03]	+0.011 [−0.03, +0.06]	+0.031 [−0.01, +0.07]
Classic	0.539	+0.005 [−0.02, +0.03]	0 (ex.)	+0.031 [−0.04, +0.10]	0 (ex.)	+0.001 [−0.00, +0.01]	−0.000 [−0.01, +0.01]	−0.005 [−0.02, +0.02]	−0.001 [−0.03, +0.03]
Dur. trend	0.549	+0.014 [−0.04, +0.07]	+0.016 [−0.03, +0.06]	+0.036 [−0.05, +0.12]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.576	**+0.093 [+0.01, +0.17]**	0 (ex.)	0 (ex.)	0 (ex.)	+0.037 [−0.05, +0.12]	**+0.071 [+0.02, +0.12]**	+0.072 [−0.01, +0.15]	+0.062 [−0.03, +0.16]
RQA 1u	0.521	+0.047 [−0.03, +0.12]	0 (ex.)	0 (ex.)	0 (ex.)	−0.007 [−0.05, +0.04]	+0.017 [−0.06, +0.09]	+0.028 [−0.06, +0.12]	+0.007 [−0.08, +0.09]
Directed	0.531	+0.028 [−0.05, +0.11]	0 (ex.)	0 (ex.)	+0.007 [−0.06, +0.07]	+0.003 [−0.06, +0.06]	+0.020 [−0.03, +0.07]	+0.038 [−0.02, +0.10]	+0.039 [−0.03, +0.11]
FAG lattice	0.591	0 (ex.)	+0.010 [−0.01, +0.03]	+0.014 [−0.01, +0.04]	+0.042 [−0.03, +0.11]	+0.036 [−0.04, +0.11]	+0.056 [−0.03, +0.14]	+0.067 [−0.02, +0.16]	+0.081 [−0.01, +0.17]

Table S6: Profile of `sb_native`: native vs. non-native English readers, reading pages. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.624	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.545	0 (ex.)	0 (ex.)	0 (ex.)	+0.005 [−0.06, +0.07]	+0.000 [−0.01, +0.01]	+0.000 [−0.02, +0.02]	−0.003 [−0.04, +0.03]	+0.006 [−0.06, +0.07]
Classic	0.689	+0.037 [−0.08, +0.15]	0 (ex.)	+0.017 [−0.07, +0.10]	0 (ex.)	+0.001 [−0.00, +0.00]	+0.004 [−0.00, +0.01]	+0.012 [−0.01, +0.03]	+0.022 [−0.04, +0.08]
Dur. trend	0.654	+0.011 [−0.04, +0.06]	+0.012 [−0.03, +0.06]	**+0.144 [+0.00, +0.29]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.614	+0.074 [−0.01, +0.16]	0 (ex.)	0 (ex.)	0 (ex.)	+0.012 [−0.02, +0.04]	+0.029 [−0.01, +0.07]	**+0.079 [+0.01, +0.15]**	**+0.088 [+0.01, +0.16]**
RQA 1u	0.669	+0.105 [−0.00, +0.21]	0 (ex.)	0 (ex.)	0 (ex.)	−0.006 [−0.03, +0.02]	+0.015 [−0.02, +0.05]	**+0.086 [+0.02, +0.16]**	**+0.140 [+0.05, +0.23]**
Directed	0.614	−0.030 [−0.09, +0.03]	0 (ex.)	0 (ex.)	−0.004 [−0.04, +0.03]	−0.011 [−0.04, +0.02]	−0.015 [−0.06, +0.03]	−0.014 [−0.06, +0.04]	−0.004 [−0.07, +0.06]
FAG lattice	0.587	0 (ex.)	−0.000 [−0.02, +0.02]	+0.008 [−0.02, +0.03]	−0.002 [−0.04, +0.03]	−0.004 [−0.03, +0.02]	−0.011 [−0.06, +0.03]	−0.018 [−0.07, +0.04]	−0.022 [−0.09, +0.04]

Table S7: Profile of `sb_compr`: high vs. low passage comprehension, reading pages. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.554	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.413	0 (ex.)	0 (ex.)	0 (ex.)	−0.047 [−0.12, +0.03]	−0.004 [−0.02, +0.01]	−0.007 [−0.03, +0.01]	**−0.035 [−0.06, −0.01]**	−0.041 [−0.11, +0.03]
Classic	0.587	+0.010 [−0.07, +0.09]	0 (ex.)	+0.026 [−0.09, +0.14]	0 (ex.)	+0.000 [−0.00, +0.00]	+0.001 [−0.01, +0.01]	+0.000 [−0.01, +0.01]	+0.003 [−0.04, +0.05]
Dur. trend	0.575	−0.006 [−0.04, +0.03]	−0.007 [−0.03, +0.02]	+0.070 [−0.05, +0.19]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.497	+0.011 [−0.12, +0.14]	0 (ex.)	0 (ex.)	0 (ex.)	+0.009 [−0.06, +0.07]	−0.003 [−0.10, +0.09]	+0.001 [−0.08, +0.08]	−0.005 [−0.11, +0.09]
RQA 1u	0.529	+0.050 [−0.04, +0.14]	0 (ex.)	0 (ex.)	0 (ex.)	−0.017 [−0.06, +0.03]	−0.014 [−0.07, +0.04]	+0.006 [−0.05, +0.07]	+0.032 [−0.08, +0.15]
Directed	0.533	+0.004 [−0.07, +0.08]	0 (ex.)	0 (ex.)	−0.010 [−0.10, +0.07]	+0.013 [−0.05, +0.07]	−0.004 [−0.08, +0.07]	+0.003 [−0.07, +0.08]	+0.010 [−0.07, +0.09]
FAG lattice	0.506	0 (ex.)	−0.007 [−0.04, +0.02]	−0.002 [−0.03, +0.03]	−0.048 [−0.10, +0.00]	−0.032 [−0.10, +0.04]	−0.035 [−0.11, +0.04]	−0.038 [−0.13, +0.05]	−0.011 [−0.10, +0.08]

Table S8: Profile of `em_program`: Rectangle vs. Vehicle program. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.561	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.957	0 (ex.)	0 (ex.)	0 (ex.)	**+0.139 [+0.08, +0.20]**	−0.001 [−0.01, +0.00]	−0.001 [−0.01, +0.01]	+0.001 [−0.01, +0.01]	+0.002 [−0.01, +0.02]
Classic	0.539	**−0.249 [−0.34, −0.16]**	0 (ex.)	+0.032 [−0.08, +0.15]	0 (ex.)	−0.000 [−0.02, +0.02]	+0.003 [−0.03, +0.04]	−0.004 [−0.06, +0.05]	+0.008 [−0.10, +0.11]
Dur. trend	0.540	+0.017 [−0.13, +0.16]	+0.038 [−0.12, +0.20]	+0.055 [−0.07, +0.18]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.580	+0.036 [−0.07, +0.14]	0 (ex.)	0 (ex.)	0 (ex.)	+0.004 [−0.09, +0.09]	+0.061 [−0.08, +0.21]	+0.068 [−0.12, +0.25]	+0.000 [−0.17, +0.17]
RQA 1u	0.547	+0.003 [−0.15, +0.16]	0 (ex.)	0 (ex.)	0 (ex.)	−0.010 [−0.10, +0.08]	+0.003 [−0.13, +0.13]	−0.046 [−0.22, +0.13]	−0.067 [−0.26, +0.12]
Directed	0.655	−0.003 [−0.08, +0.07]	0 (ex.)	0 (ex.)	+0.090 [−0.00, +0.18]	+0.112 [−0.02, +0.24]	+0.141 [−0.01, +0.29]	+0.035 [−0.08, +0.15]	+0.044 [−0.10, +0.19]
FAG lattice	0.658	0 (ex.)	−0.002 [−0.03, +0.03]	+0.001 [−0.03, +0.03]	+0.025 [−0.12, +0.17]	+0.022 [−0.04, +0.08]	−0.049 [−0.14, +0.04]	−0.032 [−0.17, +0.11]	+0.005 [−0.14, +0.15]
FAG AOI	0.778	0 (ex.)	−0.005 [−0.02, +0.01]	−0.005 [−0.02, +0.01]	+0.079 [−0.04, +0.20]	−0.022 [−0.08, +0.04]	−0.013 [−0.10, +0.07]	−0.021 [−0.14, +0.09]	−0.044 [−0.14, +0.05]
FAG AOI pad.	0.729	0 (ex.)	+0.001 [−0.02, +0.03]	+0.001 [−0.03, +0.03]	+0.056 [−0.06, +0.17]	−0.023 [−0.10, +0.06]	+0.017 [−0.09, +0.12]	+0.028 [−0.08, +0.13]	+0.049 [−0.09, +0.19]

Table S9: Profile of `em_halves`: first vs. second half of a program reading. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.500	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.616	0 (ex.)	0 (ex.)	0 (ex.)	+0.012 [−0.05, +0.07]	−0.001 [−0.01, +0.01]	+0.000 [−0.01, +0.01]	+0.001 [−0.02, +0.02]	+0.008 [−0.02, +0.04]
Classic	0.776	**+0.172 [+0.11, +0.23]**	0 (ex.)	+0.004 [−0.01, +0.02]	0 (ex.)	−0.004 [−0.01, +0.00]	−0.003 [−0.02, +0.01]	−0.002 [−0.03, +0.02]	+0.043 [−0.00, +0.09]
Dur. trend	0.582	+0.048 [−0.04, +0.14]	+0.036 [−0.08, +0.15]	**+0.086 [+0.00, +0.17]**	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.613	+0.058 [−0.04, +0.15]	0 (ex.)	0 (ex.)	0 (ex.)	−0.036 [−0.09, +0.02]	−0.050 [−0.11, +0.01]	**−0.085 [−0.15, −0.02]**	−0.053 [−0.11, +0.01]
RQA 1u	0.680	+0.081 [−0.00, +0.16]	0 (ex.)	0 (ex.)	0 (ex.)	−0.006 [−0.02, +0.01]	−0.018 [−0.05, +0.01]	**−0.052 [−0.10, −0.01]**	−0.043 [−0.09, +0.01]
Directed	0.634	+0.033 [−0.04, +0.10]	0 (ex.)	0 (ex.)	−0.004 [−0.06, +0.05]	−0.010 [−0.07, +0.05]	−0.010 [−0.07, +0.05]	+0.002 [−0.07, +0.08]	+0.059 [−0.03, +0.15]
FAG lattice	0.599	0 (ex.)	−0.008 [−0.03, +0.01]	−0.006 [−0.03, +0.01]	−0.010 [−0.07, +0.05]	−0.007 [−0.07, +0.05]	−0.000 [−0.08, +0.07]	+0.006 [−0.07, +0.08]	+0.036 [−0.08, +0.15]
FAG AOI	0.682	0 (ex.)	−0.000 [−0.01, +0.01]	−0.002 [−0.01, +0.01]	**+0.098 [+0.03, +0.17]**	+0.036 [−0.03, +0.10]	+0.023 [−0.02, +0.07]	+0.045 [−0.03, +0.12]	**+0.113 [+0.01, +0.21]**
FAG AOI pad.	0.669	0 (ex.)	−0.004 [−0.02, +0.02]	−0.001 [−0.02, +0.02]	+0.047 [−0.02, +0.11]	+0.027 [−0.04, +0.09]	+0.016 [−0.06, +0.09]	+0.016 [−0.05, +0.09]	+0.062 [−0.02, +0.14]

Table S10: Profile of `em_expertise`: high vs. low/no self-rated programming expertise. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.472	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.469	0 (ex.)	0 (ex.)	0 (ex.)	−0.028 [−0.21, +0.15]	+0.006 [−0.03, +0.04]	+0.004 [−0.05, +0.06]	+0.007 [−0.06, +0.08]	−0.004 [−0.10, +0.09]
Classic	0.438	−0.032 [−0.28, +0.21]	0 (ex.)	−0.023 [−0.34, +0.29]	0 (ex.)	−0.002 [−0.04, +0.03]	−0.008 [−0.05, +0.03]	−0.043 [−0.10, +0.01]	−0.031 [−0.16, +0.09]
Dur. trend	0.541	+0.046 [−0.13, +0.23]	+0.070 [−0.23, +0.37]	+0.031 [−0.27, +0.33]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.419	−0.016 [−0.37, +0.34]	0 (ex.)	0 (ex.)	0 (ex.)	−0.022 [−0.21, +0.17]	−0.033 [−0.11, +0.04]	+0.068 [−0.13, +0.27]	+0.056 [−0.21, +0.32]
RQA 1u	0.416	−0.037 [−0.37, +0.30]	0 (ex.)	0 (ex.)	0 (ex.)	−0.017 [−0.13, +0.09]	+0.016 [−0.24, +0.28]	+0.028 [−0.21, +0.27]	+0.090 [−0.24, +0.42]
Directed	0.366	+0.040 [−0.12, +0.20]	0 (ex.)	0 (ex.)	−0.018 [−0.10, +0.06]	+0.044 [−0.17, +0.26]	+0.013 [−0.25, +0.28]	−0.086 [−0.31, +0.14]	−0.083 [−0.32, +0.15]
FAG lattice	0.447	0 (ex.)	+0.022 [−0.03, +0.08]	+0.028 [−0.07, +0.12]	−0.026 [−0.25, +0.20]	−0.035 [−0.28, +0.21]	−0.065 [−0.28, +0.15]	−0.062 [−0.39, +0.27]	−0.058 [−0.29, +0.18]
FAG AOI	0.259	0 (ex.)	−0.009 [−0.07, +0.06]	−0.017 [−0.11, +0.07]	−0.231 [−0.48, +0.02]	−0.128 [−0.35, +0.09]	−0.143 [−0.37, +0.09]	−0.196 [−0.49, +0.10]	−0.110 [−0.39, +0.17]
FAG AOI pad.	0.459	0 (ex.)	+0.015 [−0.05, +0.08]	+0.014 [−0.05, +0.07]	−0.034 [−0.26, +0.20]	+0.062 [−0.20, +0.33]	+0.010 [−0.25, +0.27]	−0.010 [−0.27, +0.25]	−0.023 [−0.30, +0.26]

Table S11: Profile of `em_correct`: correct vs. incorrect answer, both programs pooled (confounded). Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.495	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.716	0 (ex.)	0 (ex.)	0 (ex.)	**+0.096 [+0.01, +0.18]**	+0.000 [−0.01, +0.01]	+0.003 [−0.01, +0.01]	−0.001 [−0.01, +0.01]	+0.005 [−0.03, +0.04]
Classic	0.501	−0.123 [−0.27, +0.03]	0 (ex.)	+0.054 [−0.07, +0.18]	0 (ex.)	−0.000 [−0.02, +0.02]	−0.000 [−0.03, +0.03]	−0.004 [−0.06, +0.05]	−0.002 [−0.07, +0.07]
Dur. trend	0.511	−0.013 [−0.17, +0.15]	−0.023 [−0.17, +0.13]	+0.046 [−0.09, +0.18]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.571	+0.022 [−0.10, +0.14]	0 (ex.)	0 (ex.)	0 (ex.)	+0.040 [−0.05, +0.13]	+0.056 [−0.05, +0.16]	+0.097 [−0.06, +0.25]	+0.061 [−0.12, +0.24]
RQA 1u	0.525	−0.031 [−0.19, +0.13]	0 (ex.)	0 (ex.)	0 (ex.)	+0.018 [−0.05, +0.09]	+0.069 [−0.04, +0.18]	+0.062 [−0.08, +0.20]	+0.009 [−0.18, +0.20]
Directed	0.580	−0.021 [−0.11, +0.07]	0 (ex.)	0 (ex.)	+0.043 [−0.05, +0.14]	+0.053 [−0.04, +0.15]	+0.103 [−0.02, +0.23]	+0.095 [−0.05, +0.24]	+0.058 [−0.12, +0.23]
FAG lattice	0.664	0 (ex.)	+0.001 [−0.01, +0.02]	−0.002 [−0.02, +0.01]	+0.002 [−0.07, +0.08]	+0.002 [−0.07, +0.07]	−0.004 [−0.08, +0.08]	+0.051 [−0.07, +0.17]	+0.054 [−0.07, +0.18]
FAG AOI	0.635	0 (ex.)	−0.004 [−0.03, +0.02]	−0.001 [−0.02, +0.02]	+0.076 [−0.09, +0.24]	+0.014 [−0.09, +0.12]	+0.052 [−0.06, +0.17]	−0.008 [−0.15, +0.13]	+0.031 [−0.11, +0.17]
FAG AOI pad.	0.571	0 (ex.)	−0.006 [−0.03, +0.02]	−0.010 [−0.04, +0.02]	+0.052 [−0.09, +0.19]	−0.020 [−0.16, +0.12]	−0.022 [−0.16, +0.12]	−0.007 [−0.15, +0.14]	+0.004 [−0.15, +0.16]

Table S12: Profile of `em_correctrect`: correct vs. incorrect answer, Rectangle program only. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.589	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.472	0 (ex.)	0 (ex.)	0 (ex.)	−0.062 [−0.25, +0.13]	+0.002 [−0.02, +0.02]	+0.006 [−0.03, +0.04]	+0.034 [−0.02, +0.09]	+0.073 [−0.02, +0.17]
Classic	0.523	−0.002 [−0.19, +0.19]	0 (ex.)	+0.048 [−0.19, +0.28]	0 (ex.)	+0.004 [−0.03, +0.04]	+0.008 [−0.05, +0.06]	−0.006 [−0.08, +0.06]	+0.003 [−0.08, +0.09]
Dur. trend	0.507	−0.069 [−0.23, +0.09]	−0.077 [−0.24, +0.08]	−0.037 [−0.29, +0.22]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.451	−0.054 [−0.38, +0.27]	0 (ex.)	0 (ex.)	0 (ex.)	−0.046 [−0.25, +0.16]	−0.072 [−0.22, +0.08]	−0.004 [−0.26, +0.25]	−0.041 [−0.28, +0.20]
RQA 1u	0.462	−0.065 [−0.38, +0.25]	0 (ex.)	0 (ex.)	0 (ex.)	−0.015 [−0.15, +0.12]	−0.026 [−0.18, +0.13]	−0.046 [−0.24, +0.15]	−0.048 [−0.29, +0.20]
Directed	0.533	+0.006 [−0.22, +0.23]	0 (ex.)	0 (ex.)	−0.013 [−0.19, +0.17]	+0.000 [−0.18, +0.19]	+0.014 [−0.19, +0.22]	−0.000 [−0.22, +0.22]	+0.018 [−0.19, +0.23]
FAG lattice	0.578	0 (ex.)	+0.007 [−0.05, +0.06]	+0.011 [−0.04, +0.06]	+0.021 [−0.11, +0.16]	+0.040 [−0.15, +0.23]	+0.025 [−0.11, +0.16]	+0.057 [−0.18, +0.29]	+0.097 [−0.06, +0.25]
FAG AOI	0.658	0 (ex.)	+0.008 [−0.05, +0.07]	+0.005 [−0.06, +0.07]	+0.186 [−0.03, +0.40]	+0.092 [−0.16, +0.34]	**+0.167 [+0.01, +0.32]**	+0.144 [−0.08, +0.37]	+0.201 [−0.02, +0.42]
FAG AOI pad.	0.544	0 (ex.)	+0.024 [−0.03, +0.08]	+0.020 [−0.05, +0.09]	+0.029 [−0.17, +0.23]	−0.027 [−0.19, +0.13]	−0.003 [−0.23, +0.22]	+0.069 [−0.18, +0.32]	+0.044 [−0.19, +0.28]

Table S13: Profile of `em_correctveh`: correct vs. incorrect answer, Vehicle program only. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

Descriptor	AUC	order	dur. coupl.	dur. marg.	rotation	jit. 0.25u	jit. 0.5u	jit. 1u	jit. 2u
Count	0.444	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
Dispersion	0.476	0 (ex.)	0 (ex.)	0 (ex.)	+0.091 [−0.20, +0.38]	+0.006 [−0.02, +0.03]	+0.007 [−0.04, +0.06]	+0.015 [−0.04, +0.07]	+0.022 [−0.10, +0.15]
Classic	0.500	+0.070 [−0.15, +0.29]	0 (ex.)	+0.025 [−0.20, +0.25]	0 (ex.)	+0.000 [−0.04, +0.04]	+0.001 [−0.06, +0.06]	+0.003 [−0.05, +0.06]	+0.017 [−0.10, +0.14]
Dur. trend	0.521	+0.012 [−0.22, +0.24]	−0.024 [−0.26, +0.21]	+0.002 [−0.30, +0.31]	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)	0 (ex.)
RQA 0.5u	0.518	−0.026 [−0.22, +0.17]	0 (ex.)	0 (ex.)	0 (ex.)	+0.034 [−0.15, +0.21]	+0.026 [−0.23, +0.28]	+0.104 [−0.14, +0.35]	+0.098 [−0.11, +0.31]
RQA 1u	0.559	+0.036 [−0.24, +0.31]	0 (ex.)	0 (ex.)	0 (ex.)	−0.004 [−0.11, +0.10]	+0.093 [−0.10, +0.29]	+0.106 [−0.09, +0.30]	+0.140 [−0.07, +0.35]
Directed	0.618	+0.134 [−0.06, +0.33]	0 (ex.)	0 (ex.)	+0.080 [−0.14, +0.31]	+0.097 [−0.11, +0.30]	+0.120 [−0.05, +0.29]	+0.147 [−0.07, +0.37]	+0.088 [−0.15, +0.33]
FAG lattice	0.574	0 (ex.)	+0.023 [−0.02, +0.07]	+0.021 [−0.05, +0.09]	+0.074 [−0.18, +0.33]	+0.066 [−0.13, +0.26]	+0.032 [−0.20, +0.26]	+0.069 [−0.17, +0.31]	+0.054 [−0.18, +0.29]
FAG AOI	0.503	0 (ex.)	+0.000 [−0.06, +0.06]	−0.004 [−0.07, +0.06]	−0.010 [−0.29, +0.27]	−0.011 [−0.24, +0.22]	−0.017 [−0.30, +0.26]	+0.014 [−0.20, +0.23]	+0.001 [−0.26, +0.27]
FAG AOI pad.	0.470	0 (ex.)	−0.007 [−0.05, +0.03]	−0.007 [−0.05, +0.04]	−0.148 [−0.43, +0.13]	+0.010 [−0.22, +0.24]	−0.052 [−0.26, +0.16]	+0.003 [−0.24, +0.24]	+0.015 [−0.24, +0.27]

Table S14: Observed pooled AUC (all out-of-fold decision values ranked together; class-balanced participant folds), with the permutation p-value in brackets (200 permutations under the schemes of Table 2 of the article).

Problem	Count	Dispersion	Classic	Dur. trend	RQA $0.5u$	RQA $1u$	Directed	FAG lattice	FAG AOI	FAG AOI pad.
`gb_tex_vd1`	0.968 (.005)	1.000 (.005)	0.968 (.005)	0.956 (.005)	0.999 (.005)	1.000 (.005)	0.977 (.005)	0.989 (.005)	–	–
`gb_vd1_blg`	0.691 (.005)	1.000 (.005)	0.888 (.005)	0.751 (.005)	0.999 (.005)	1.000 (.005)	1.000 (.005)	0.957 (.005)	–	–
`gb_vd1_vd2`	0.633 (.005)	0.703 (.005)	0.636 (.005)	0.791 (.005)	0.672 (.005)	0.715 (.005)	0.700 (.005)	0.666 (.005)	–	–
`gb_vd1_halves`	0.500 (1.000)	0.921 (.005)	0.561 (.005)	0.590 (.005)	0.742 (.005)	0.822 (.005)	0.803 (.005)	0.739 (.005)	–	–
`gb_tex_s1s2`	0.516 (.065)	0.517 (.224)	0.537 (.005)	0.545 (.015)	0.574 (.005)	0.515 (.159)	0.531 (.070)	0.588 (.005)	–	–
`sb_native`	0.610 (.010)	0.530 (.164)	0.673 (.005)	0.637 (.015)	0.611 (.005)	0.659 (.005)	0.594 (.025)	0.575 (.050)	–	–
`sb_compr`	0.527 (.915)	0.402 (.990)	0.595 (.950)	0.575 (.910)	0.485 (.896)	0.523 (.891)	0.514 (.935)	0.473 (.955)	–	–
`em_program`	0.547 (.025)	0.963 (.005)	0.546 (.025)	0.524 (.249)	0.587 (.015)	0.541 (.129)	0.653 (.005)	0.666 (.005)	0.781 (.005)	0.724 (.005)
`em_halves`	0.500 (1.000)	0.619 (.005)	0.768 (.005)	0.574 (.010)	0.606 (.005)	0.677 (.005)	0.629 (.005)	0.602 (.005)	0.675 (.005)	0.664 (.005)
`em_expertise`	0.400 (.697)	0.449 (.637)	0.378 (.816)	0.510 (.373)	0.425 (.771)	0.371 (.881)	0.319 (.940)	0.414 (.816)	0.264 (.985)	0.418 (.756)
`em_correct`	0.463 (.741)	0.710 (.478)	0.497 (.522)	0.505 (.393)	0.554 (.179)	0.518 (.328)	0.559 (.279)	0.659 (.015)	0.638 (.254)	0.569 (.746)
`em_correctrect`	0.539 (.214)	0.446 (.741)	0.488 (.557)	0.530 (.303)	0.455 (.692)	0.452 (.726)	0.518 (.368)	0.556 (.184)	0.634 (.035)	0.514 (.403)
`em_correctveh`	0.434 (.701)	0.445 (.731)	0.458 (.672)	0.474 (.627)	0.490 (.507)	0.564 (.159)	0.588 (.119)	0.536 (.299)	0.465 (.592)	0.465 (.627)

Table S15: Range, over descriptors, of the mean of the permutation distribution of the observed AUC, under the within-fold AUC (1000 permutations) and the pooled AUC (200 permutations), both with class-balanced folds. On `em_correct` labels are permuted within programs and on `sb_compr` between the passages of a participant, so stimulus structure that survives the permutation raises the mean for descriptors that read it.

Problem	Within-fold AUC	Pooled AUC
`gb_tex_vd1`	0.499–0.501	0.491–0.496
`gb_vd1_blg`	0.499–0.502	0.494–0.497
`gb_vd1_vd2`	0.499–0.501	0.494–0.499
`gb_vd1_halves`	0.499–0.501	0.496–0.500
`gb_tex_s1s2`	0.499–0.501	0.494–0.498
`sb_native`	0.497–0.504	0.476–0.490
`sb_compr`	0.477–0.618	0.458–0.611
`em_program`	0.496–0.501	0.488–0.500
`em_halves`	0.498–0.502	0.487–0.500
`em_expertise`	0.489–0.504	0.466–0.492
`em_correct`	0.502–0.710	0.487–0.708
`em_correctrect`	0.496–0.505	0.476–0.492
`em_correctveh`	0.496–0.503	0.475–0.491

Table S16: Evidence ledger of the 89 interpreted cells. Primary: sensitivity with the within-fold AUC and class-balanced folds ($R = 6$, $B = 60$, seed 42), with its 95% refit-bootstrap interval (excluding zero by selection). Pooled AUC: the same cell with the pooled AUC. Partitions: range of the point estimate over 20 fold partitions. Long runs: range of the point estimate over three runs with $R = 20$ and $B = 300$ (seeds 42, 43, 44), and the number of those runs whose interval excluded zero. Trees: gradient-boosted trees ($R = 6$, $B = 60$, seed 42). Norm.: sensitivity divided by the observed AUC minus the mean of its permutation distribution, under the within-fold and the pooled AUC. Robust: sign and exclusion of zero hold in the primary run, the pooled run, every partition and every long run and, for the AOI graph, on the padded graph (Table S23 and Section 5.9 of the article). Bold: interval excludes zero.

Problem	Descriptor	Transformation	Primary	Pooled AUC	Partitions	Long runs	Trees	Norm.	Norm. (pooled)	Robust
gb_tex_vd1	Classic	duration marginal	+0.035 [+0.021, +0.048]	**+0.035 [+0.022, +0.048]**	+0.033 to +0.036	+0.034 to +0.034 (3/3)	**+0.045 [+0.025, +0.065]**	+0.07	+0.07	✓
gb_tex_vd1	Classic	order	−0.022 [−0.034, −0.010]	**−0.022 [−0.034, −0.009]**	−0.024 to −0.020	−0.022 to −0.022 (3/3)	**−0.019 [−0.033, −0.005]**	−0.05	−0.05	✓
gb_tex_vd1	Classic	jitter 0.25u	−0.003 [−0.005, −0.001]	**−0.003 [−0.005, −0.001]**	−0.003 to −0.002	−0.003 to −0.003 (3/3)	−0.002 [−0.009, +0.004]	−0.01	−0.01	✓
gb_tex_vd1	Classic	jitter 0.5u	−0.006 [−0.009, −0.002]	**−0.006 [−0.008, −0.003]**	−0.007 to −0.005	−0.006 to −0.005 (3/3)	−0.005 [−0.011, +0.001]	−0.01	−0.01	✓
gb_tex_vd1	Classic	jitter 1u	−0.007 [−0.011, −0.003]	**−0.007 [−0.011, −0.003]**	−0.009 to −0.007	−0.007 to −0.007 (3/3)	−0.006 [−0.014, +0.003]	−0.01	−0.02	✓
gb_tex_vd1	Directed	order	−0.014 [−0.020, −0.008]	**−0.014 [−0.020, −0.008]**	−0.014 to −0.012	−0.015 to −0.014 (3/3)	−0.004 [−0.011, +0.002]	−0.03	−0.03	✓
gb_tex_vd1	Directed	jitter 0.25u	−0.011 [−0.018, −0.004]	**−0.011 [−0.016, −0.005]**	−0.013 to −0.011	−0.012 to −0.011 (3/3)	−0.004 [−0.010, +0.002]	−0.02	−0.02	✓
gb_tex_vd1	Directed	jitter 0.5u	−0.015 [−0.021, −0.009]	**−0.015 [−0.020, −0.009]**	−0.015 to −0.013	−0.016 to −0.015 (3/3)	−0.005 [−0.011, +0.001]	−0.03	−0.03	✓
gb_tex_vd1	Directed	jitter 1u	−0.016 [−0.022, −0.011]	**−0.016 [−0.022, −0.010]**	−0.017 to −0.014	−0.017 to −0.016 (3/3)	−0.006 [−0.012, +0.000]	−0.03	−0.03	✓
gb_tex_vd1	Directed	jitter 2u	−0.013 [−0.021, −0.005]	**−0.012 [−0.019, −0.005]**	−0.013 to −0.010	−0.013 to −0.012 (3/3)	−0.002 [−0.011, +0.007]	−0.03	−0.03	✓
gb_tex_vd1	Directed	rotation	−0.007 [−0.013, −0.002]	**−0.008 [−0.012, −0.003]**	−0.009 to −0.007	−0.007 to −0.007 (3/3)	−0.002 [−0.009, +0.005]	−0.02	−0.02	✓
gb_tex_vd1	Dispersion	rotation	+0.021 [+0.014, +0.029]	**+0.021 [+0.014, +0.029]**	+0.018 to +0.020	+0.021 to +0.021 (3/3)	**+0.016 [+0.008, +0.024]**	+0.04	+0.04	✓
gb_tex_vd1	Dur. trend	duration marginal	+0.473 [+0.442, +0.503]	**+0.474 [+0.446, +0.503]**	+0.466 to +0.479	+0.454 to +0.469 (3/3)	**+0.371 [+0.318, +0.424]**	+1.03	+1.03	✓
gb_tex_vd1	FAG lattice	jitter 1u	+0.009 [+0.002, +0.016]	**+0.010 [+0.002, +0.017]**	+0.007 to +0.009	+0.007 to +0.009 (3/3)	+0.005 [−0.003, +0.013]	+0.02	+0.02	✓
gb_tex_vd1	FAG lattice	jitter 2u	+0.049 [+0.037, +0.061]	**+0.050 [+0.038, +0.062]**	+0.044 to +0.047	+0.047 to +0.048 (3/3)	**+0.041 [+0.024, +0.058]**	+0.10	+0.10	✓
gb_tex_vd1	RQA 0.5u	order	+0.010 [+0.003, +0.018]	**+0.011 [+0.004, +0.018]**	+0.009 to +0.011	+0.010 to +0.010 (3/3)	**+0.013 [+0.004, +0.022]**	+0.02	+0.02	✓
gb_tex_vd1	RQA 0.5u	jitter 1u	+0.004 [+0.001, +0.008]	**+0.004 [+0.001, +0.008]**	+0.003 to +0.004	+0.003 to +0.004 (3/3)	+0.004 [−0.000, +0.007]	+0.01	+0.01	✓
gb_tex_vd1	RQA 0.5u	jitter 2u	+0.015 [+0.009, +0.021]	**+0.016 [+0.010, +0.021]**	+0.017 to +0.019	+0.017 to +0.018 (3/3)	**+0.014 [+0.009, +0.020]**	+0.03	+0.03	✓
gb_tex_vd1	RQA 1u	order	+0.006 [+0.001, +0.012]	**+0.007 [+0.001, +0.012]**	+0.006 to +0.007	+0.006 to +0.006 (3/3)	**+0.007 [+0.001, +0.013]**	+0.01	+0.01	✓
gb_tex_vd1	RQA 1u	jitter 2u	+0.006 [+0.001, +0.011]	**+0.007 [+0.002, +0.011]**	+0.006 to +0.007	+0.006 to +0.006 (3/3)	**+0.007 [+0.001, +0.012]**	+0.01	+0.01	✓
gb_vd1_blg	Classic	duration marginal	+0.033 [+0.020, +0.046]	**+0.034 [+0.022, +0.047]**	+0.030 to +0.036	+0.033 to +0.033 (3/3)	**+0.037 [+0.009, +0.065]**	+0.09	+0.09	✓
gb_vd1_blg	Classic	order	−0.114 [−0.137, −0.091]	**−0.112 [−0.133, −0.091]**	−0.114 to −0.110	−0.114 to −0.114 (3/3)	**−0.118 [−0.148, −0.088]**	−0.30	−0.29	✓
gb_vd1_blg	Classic	jitter 0.25u	−0.003 [−0.004, −0.001]	**−0.002 [−0.004, −0.001]**	−0.003 to −0.002	−0.003 to −0.003 (3/3)	+0.001 [−0.008, +0.009]	−0.01	−0.01	✓
gb_vd1_blg	Classic	jitter 0.5u	−0.007 [−0.010, −0.005]	**−0.007 [−0.009, −0.005]**	−0.007 to −0.006	−0.007 to −0.007 (3/3)	−0.003 [−0.012, +0.006]	−0.02	−0.02	✓
gb_vd1_blg	Classic	jitter 1u	−0.014 [−0.021, −0.007]	**−0.014 [−0.020, −0.008]**	−0.015 to −0.013	−0.014 to −0.014 (3/3)	−0.009 [−0.021, +0.003]	−0.04	−0.04	✓
gb_vd1_blg	Classic	jitter 2u	−0.014 [−0.023, −0.005]	**−0.013 [−0.021, −0.006]**	−0.015 to −0.013	−0.016 to −0.014 (3/3)	−0.005 [−0.023, +0.012]	−0.04	−0.03	✓
gb_vd1_blg	Directed	order	+0.005 [+0.001, +0.008]	**+0.005 [+0.002, +0.008]**	+0.004 to +0.005	+0.005 to +0.005 (3/3)	**+0.006 [+0.002, +0.010]**	+0.01	+0.01	✓
gb_vd1_blg	Directed	jitter 2u	+0.002 [+0.000, +0.004]	**+0.002 [+0.001, +0.004]**	+0.002 to +0.002	+0.002 to +0.002 (2/3)	+0.003 [−0.000, +0.006]	+0.00	+0.00	–
gb_vd1_blg	Dur. trend	duration marginal	+0.252 [+0.199, +0.304]	**+0.257 [+0.206, +0.308]**	+0.251 to +0.265	+0.243 to +0.259 (3/3)	**+0.226 [+0.161, +0.291]**	+1.01	+1.00	✓
gb_vd1_blg	FAG lattice	duration marginal	−0.008 [−0.013, −0.004]	**−0.009 [−0.013, −0.005]**	−0.009 to −0.008	−0.009 to −0.008 (3/3)	−0.004 [−0.009, +0.002]	−0.02	−0.02	✓
gb_vd1_blg	FAG lattice	jitter 1u	+0.019 [+0.003, +0.035]	**+0.020 [+0.004, +0.035]**	+0.017 to +0.021	+0.017 to +0.019 (3/3)	+0.011 [−0.000, +0.022]	+0.04	+0.04	✓
gb_vd1_blg	FAG lattice	jitter 2u	+0.027 [+0.010, +0.044]	**+0.027 [+0.011, +0.043]**	+0.028 to +0.030	+0.026 to +0.027 (3/3)	**+0.024 [+0.009, +0.039]**	+0.06	+0.06	✓
gb_vd1_blg	FAG lattice	rotation	+0.029 [+0.016, +0.042]	**+0.029 [+0.018, +0.041]**	+0.024 to +0.028	+0.024 to +0.028 (3/3)	**+0.023 [+0.009, +0.036]**	+0.06	+0.06	✓
gb_vd1_blg	RQA 0.5u	order	+0.013 [+0.006, +0.020]	**+0.013 [+0.007, +0.019]**	+0.012 to +0.014	+0.013 to +0.013 (3/3)	**+0.014 [+0.005, +0.023]**	+0.03	+0.03	✓
gb_vd1_blg	RQA 0.5u	jitter 2u	+0.003 [+0.001, +0.006]	**+0.003 [+0.001, +0.006]**	+0.005 to +0.006	+0.004 to +0.004 (3/3)	**+0.006 [+0.001, +0.010]**	+0.01	+0.01	✓
gb_vd1_blg	RQA 1u	order	+0.001 [+0.000, +0.002]	**+0.001 [+0.000, +0.002]**	+0.001 to +0.001	+0.001 to +0.001 (3/3)	+0.002 [−0.001, +0.005]	+0.00	+0.00	✓
gb_vd1_vd2	Classic	duration marginal	+0.124 [+0.099, +0.150]	**+0.124 [+0.103, +0.144]**	+0.124 to +0.135	+0.122 to +0.127 (3/3)	**+0.130 [+0.073, +0.187]**	+0.88	+0.90	✓
gb_vd1_vd2	Classic	jitter 0.5u	+0.004 [+0.000, +0.007]	+0.002 [−0.000, +0.004]	+0.001 to +0.003	+0.003 to +0.003 (2/3)	+0.002 [−0.026, +0.031]	+0.03	+0.01	–
gb_vd1_vd2	Directed	order	+0.040 [+0.015, +0.065]	**+0.042 [+0.018, +0.065]**	+0.038 to +0.045	+0.037 to +0.039 (3/3)	**+0.059 [+0.007, +0.111]**	+0.20	+0.21	✓
gb_vd1_vd2	Directed	jitter 1u	+0.042 [+0.009, +0.075]	**+0.044 [+0.012, +0.076]**	+0.040 to +0.049	+0.040 to +0.042 (3/3)	+0.048 [−0.000, +0.096]	+0.21	+0.22	✓
gb_vd1_vd2	Directed	jitter 2u	+0.069 [+0.044, +0.095]	**+0.071 [+0.045, +0.097]**	+0.065 to +0.072	+0.067 to +0.068 (3/3)	**+0.086 [+0.040, +0.133]**	+0.35	+0.35	✓
gb_vd1_vd2	Dispersion	jitter 0.5u	+0.012 [+0.000, +0.023]	**+0.011 [+0.002, +0.021]**	+0.010 to +0.013	+0.012 to +0.013 (3/3)	+0.035 [−0.015, +0.086]	+0.06	+0.06	✓
gb_vd1_vd2	Dispersion	jitter 1u	+0.045 [+0.026, +0.063]	**+0.043 [+0.027, +0.060]**	+0.039 to +0.043	+0.040 to +0.048 (3/3)	**+0.069 [+0.005, +0.134]**	+0.21	+0.21	✓
gb_vd1_vd2	Dispersion	jitter 2u	+0.116 [+0.085, +0.148]	**+0.113 [+0.082, +0.143]**	+0.117 to +0.123	+0.113 to +0.120 (3/3)	**+0.102 [+0.038, +0.166]**	+0.56	+0.55	✓
gb_vd1_vd2	Dispersion	rotation	+0.098 [+0.068, +0.128]	**+0.096 [+0.069, +0.124]**	+0.108 to +0.114	+0.100 to +0.103 (3/3)	**+0.108 [+0.045, +0.170]**	+0.47	+0.47	✓
gb_vd1_vd2	Dur. trend	duration coupling	+0.158 [+0.131, +0.185]	**+0.162 [+0.140, +0.184]**	+0.154 to +0.162	+0.157 to +0.158 (3/3)	**+0.176 [+0.133, +0.219]**	+0.54	+0.55	✓
gb_vd1_vd2	Dur. trend	duration marginal	+0.286 [+0.227, +0.345]	**+0.292 [+0.233, +0.351]**	+0.280 to +0.294	+0.289 to +0.297 (3/3)	**+0.272 [+0.216, +0.329]**	+0.98	+0.98	✓
gb_vd1_vd2	Dur. trend	order	+0.157 [+0.127, +0.187]	**+0.161 [+0.134, +0.188]**	+0.150 to +0.160	+0.157 to +0.157 (3/3)	**+0.178 [+0.133, +0.224]**	+0.54	+0.54	✓
gb_vd1_vd2	FAG lattice	duration coupling	−0.015 [−0.029, −0.001]	−0.012 [−0.028, +0.003]	−0.017 to −0.012	−0.017 to −0.015 (1/3)	−0.016 [−0.046, +0.013]	−0.09	−0.07	–
gb_vd1_vd2	FAG lattice	duration marginal	−0.020 [−0.031, −0.009]	**−0.019 [−0.032, −0.007]**	−0.027 to −0.020	−0.023 to −0.020 (3/3)	−0.014 [−0.038, +0.011]	−0.12	−0.11	✓
gb_vd1_vd2	FAG lattice	jitter 2u	+0.055 [+0.013, +0.096]	**+0.058 [+0.017, +0.099]**	+0.045 to +0.053	+0.046 to +0.052 (3/3)	**+0.100 [+0.043, +0.156]**	+0.33	+0.34	✓
gb_vd1_vd2	RQA 0.5u	order	+0.035 [+0.011, +0.060]	**+0.034 [+0.013, +0.054]**	+0.032 to +0.040	+0.033 to +0.035 (3/3)	+0.044 [−0.004, +0.092]	+0.20	+0.19	✓

Problem	Descriptor	Transformation	Primary	Pooled AUC	Partitions	Long runs	Trees	Norm.	Norm. (pooled)	Robust
gb_vd1_vd2	RQA 0.5u	jitter 1u	+0.046 [+0.003, +0.090]	**+0.044 [+0.000, +0.087]**	+0.033 to +0.040	+0.035 to +0.039 (3/3)	+0.049 [−0.023, +0.121]	+0.26	+0.25	✓
gb_vd1_vd2	RQA 0.5u	jitter 2u	+0.108 [+0.071, +0.145]	**+0.106 [+0.071, +0.140]**	+0.107 to +0.114	+0.106 to +0.115 (3/3)	**+0.106 [+0.041, +0.170]**	+0.61	+0.61	✓
gb_vd1_vd2	RQA 1u	order	+0.073 [+0.043, +0.103]	**+0.073 [+0.045, +0.101]**	+0.069 to +0.076	+0.072 to +0.073 (3/3)	**+0.086 [+0.032, +0.140]**	+0.33	+0.33	✓
gb_vd1_vd2	RQA 1u	jitter 1u	+0.043 [+0.007, +0.078]	**+0.041 [+0.007, +0.075]**	+0.041 to +0.048	+0.040 to +0.042 (3/3)	+0.044 [−0.005, +0.093]	+0.19	+0.19	✓
gb_vd1_vd2	RQA 1u	jitter 2u	+0.118 [+0.080, +0.156]	**+0.115 [+0.077, +0.152]**	+0.108 to +0.115	+0.110 to +0.119 (3/3)	**+0.125 [+0.054, +0.195]**	+0.54	+0.52	✓
gb_vd1_halves	Classic	order	−0.090 [−0.110, −0.069]	**−0.091 [−0.111, −0.071]**	−0.098 to −0.088	−0.089 to −0.087 (3/3)	−0.055 [−0.114, +0.003]	−1.40	−1.45	✓
gb_vd1_halves	Directed	order	+0.036 [+0.017, +0.054]	**+0.036 [+0.019, +0.052]**	+0.032 to +0.036	+0.037 to +0.037 (3/3)	+0.027 [−0.003, +0.058]	+0.12	+0.12	✓
gb_vd1_halves	Directed	jitter 0.5u	+0.033 [+0.014, +0.052]	**+0.031 [+0.011, +0.052]**	+0.026 to +0.031	+0.031 to +0.033 (3/3)	+0.029 [−0.007, +0.065]	+0.11	+0.10	✓
gb_vd1_halves	Directed	jitter 1u	+0.090 [+0.060, +0.121]	**+0.089 [+0.061, +0.117]**	+0.087 to +0.093	+0.087 to +0.093 (3/3)	**+0.102 [+0.060, +0.145]**	+0.29	+0.29	✓
gb_vd1_halves	Directed	jitter 2u	+0.191 [+0.161, +0.220]	**+0.190 [+0.162, +0.218]**	+0.167 to +0.172	+0.186 to +0.187 (3/3)	**+0.192 [+0.140, +0.243]**	+0.62	+0.62	✓
gb_vd1_halves	Dispersion	jitter 0.5u	+0.007 [+0.002, +0.012]	**+0.007 [+0.002, +0.011]**	+0.006 to +0.008	+0.007 to +0.007 (3/3)	+0.001 [−0.011, +0.013]	+0.02	+0.02	✓
gb_vd1_halves	Dispersion	jitter 1u	+0.029 [+0.018, +0.041]	**+0.029 [+0.020, +0.038]**	+0.031 to +0.033	+0.030 to +0.032 (3/3)	**+0.028 [+0.010, +0.046]**	+0.07	+0.07	✓
gb_vd1_halves	Dispersion	jitter 2u	+0.130 [+0.111, +0.150]	**+0.129 [+0.110, +0.149]**	+0.123 to +0.126	+0.122 to +0.129 (3/3)	**+0.123 [+0.097, +0.150]**	+0.31	+0.31	✓
gb_vd1_halves	Dispersion	rotation	+0.249 [+0.224, +0.275]	**+0.247 [+0.223, +0.271]**	+0.249 to +0.253	+0.246 to +0.249 (3/3)	**+0.177 [+0.139, +0.215]**	+0.59	+0.58	✓
gb_vd1_halves	Dur. trend	duration coupling	+0.085 [+0.040, +0.130]	**+0.086 [+0.042, +0.129]**	+0.078 to +0.090	+0.078 to +0.082 (3/3)	+0.060 [−0.002, +0.121]	+0.96	+0.91	✓
gb_vd1_halves	Dur. trend	duration marginal	+0.082 [+0.044, +0.120]	**+0.088 [+0.051, +0.125]**	+0.077 to +0.085	+0.089 to +0.095 (3/3)	+0.024 [−0.051, +0.098]	+0.93	+0.94	✓
gb_vd1_halves	Dur. trend	order	+0.076 [+0.030, +0.122]	**+0.079 [+0.031, +0.126]**	+0.072 to +0.080	+0.080 to +0.081 (3/3)	+0.033 [−0.045, +0.112]	+0.86	+0.84	✓
gb_vd1_halves	FAG lattice	jitter 2u	+0.088 [+0.053, +0.123]	**+0.087 [+0.052, +0.121]**	+0.084 to +0.090	+0.082 to +0.082 (3/3)	**+0.111 [+0.049, +0.173]**	+0.37	+0.36	✓
gb_vd1_halves	RQA 0.5u	order	+0.046 [+0.023, +0.069]	**+0.046 [+0.025, +0.068]**	+0.043 to +0.049	+0.047 to +0.048 (3/3)	**+0.055 [+0.014, +0.096]**	+0.19	+0.19	✓
gb_vd1_halves	RQA 0.5u	jitter 1u	+0.079 [+0.051, +0.106]	**+0.079 [+0.051, +0.107]**	+0.068 to +0.074	+0.070 to +0.071 (3/3)	**+0.080 [+0.031, +0.129]**	+0.32	+0.33	✓
gb_vd1_halves	RQA 0.5u	jitter 2u	+0.143 [+0.097, +0.189]	**+0.143 [+0.099, +0.188]**	+0.137 to +0.142	+0.145 to +0.150 (3/3)	**+0.157 [+0.101, +0.214]**	+0.59	+0.59	✓
gb_vd1_halves	RQA 1u	order	+0.088 [+0.063, +0.112]	**+0.089 [+0.066, +0.111]**	+0.084 to +0.090	+0.085 to +0.086 (3/3)	**+0.113 [+0.077, +0.149]**	+0.27	+0.27	✓
gb_vd1_halves	RQA 1u	jitter 0.5u	+0.026 [+0.006, +0.046]	**+0.026 [+0.007, +0.045]**	+0.018 to +0.021	+0.028 to +0.031 (3/3)	+0.025 [−0.011, +0.061]	+0.08	+0.08	✓
gb_vd1_halves	RQA 1u	jitter 1u	+0.088 [+0.054, +0.121]	**+0.089 [+0.059, +0.118]**	+0.086 to +0.090	+0.089 to +0.092 (3/3)	**+0.106 [+0.071, +0.142]**	+0.27	+0.27	✓
gb_vd1_halves	RQA 1u	jitter 2u	+0.182 [+0.151, +0.214]	**+0.183 [+0.154, +0.213]**	+0.170 to +0.174	+0.179 to +0.183 (3/3)	**+0.212 [+0.172, +0.253]**	+0.56	+0.57	✓
gb_tex_s1s2	RQA 0.5u	order	+0.093 [+0.013, +0.174]	**+0.095 [+0.017, +0.173]**	+0.077 to +0.097	+0.064 to +0.083 (1/3)	+0.066 [−0.037, +0.168]	+1.23	+1.19	–
gb_tex_s1s2	RQA 0.5u	jitter 0.5u	+0.071 [+0.020, +0.123]	**+0.071 [+0.017, +0.125]**	+0.048 to +0.062	+0.057 to +0.065 (0/3)	+0.035 [−0.065, +0.135]	+0.94	+0.89	–
sb_native	RQA 0.5u	jitter 2u	+0.088 [+0.012, +0.164]	**+0.096 [+0.009, +0.182]**	+0.104 to +0.125	+0.103 to +0.117 (3/3)	−0.002 [−0.084, +0.081]	+0.79	+0.77	✓
sb_native	RQA 0.5u	jitter 1u	+0.079 [+0.007, +0.150]	+0.085 [−0.005, +0.174]	+0.055 to +0.072	+0.069 to +0.076 (2/3)	−0.020 [−0.102, +0.062]	+0.71	+0.68	–
sb_native	RQA 1u	jitter 2u	+0.140 [+0.045, +0.235]	**+0.141 [+0.035, +0.246]**	+0.139 to +0.171	+0.154 to +0.155 (3/3)	**+0.100 [+0.009, +0.190]**	+0.85	+0.81	✓
sb_native	RQA 1u	jitter 1u	+0.086 [+0.016, +0.157]	**+0.085 [+0.009, +0.161]**	+0.068 to +0.087	+0.080 to +0.084 (3/3)	+0.071 [−0.003, +0.145]	+0.52	+0.49	✓
em_program	Dispersion	rotation	+0.139 [+0.078, +0.199]	**+0.160 [+0.108, +0.213]**	+0.137 to +0.165	+0.134 to +0.142 (3/3)	**+0.166 [+0.088, +0.244]**	+0.30	+0.34	✓
em_halves	Classic	order	+0.172 [+0.112, +0.231]	**+0.171 [+0.120, +0.222]**	+0.164 to +0.191	+0.171 to +0.174 (3/3)	**+0.128 [+0.033, +0.222]**	+0.62	+0.62	✓
em_halves	FAG AOI	jitter 2u	+0.113 [+0.014, +0.212]	**+0.111 [+0.023, +0.198]**	+0.101 to +0.122	+0.111 to +0.115 (3/3)	+0.115 [−0.007, +0.237]	+0.62	+0.61	–
em_halves	FAG AOI	rotation	+0.098 [+0.031, +0.166]	**+0.093 [+0.028, +0.158]**	+0.081 to +0.100	+0.102 to +0.114 (3/3)	+0.058 [−0.060, +0.176]	+0.54	+0.51	–
em_halves	RQA 0.5u	jitter 1u	−0.085 [−0.152, −0.018]	**−0.085 [−0.145, −0.025]**	−0.089 to −0.073	−0.085 to −0.078 (3/3)	−0.089 [−0.189, +0.011]	−0.77	−0.75	✓
em_halves	RQA 1u	jitter 1u	−0.052 [−0.098, −0.006]	**−0.052 [−0.086, −0.017]**	−0.063 to −0.045	−0.050 to −0.045 (2/3)	−0.053 [−0.127, +0.020]	−0.29	−0.28	–

Table S19: Extended simulation designs ($R = 6$, $B = 60$; 40 datasets per row). Target: the reference value (60 datasets with disjoint seeds, $R = 20$) with its Monte Carlo standard error, or zero by exchangeability. Unequal: one to four trials per class and participant (W) or per participant (B); 1:3 and 16:49: class imbalance; heterogeneous: participant offset SD 100 px and participant-specific scaling.

Design	Size	Contrast	Target θ (SE)	Coverage	Excl. 0	Width	Obs. AUC
order hides spread (classic)	65 × 2 (B)	duration marginal	+0.006 (0.009)	0.975	0.025	0.397	0.520
order hides spread (classic)	65 × 2 (B)	order	−0.499 (0.010)	1.000	1.000	0.405	0.520
order hides spread (classic)	60 × 4 (W)	duration marginal	−0.001 (0.005)	1.000	0.000	0.296	0.509
order hides spread (classic)	60 × 4 (W)	order	−0.497 (0.008)	1.000	1.000	0.292	0.509
no signal (rqa)	65 × 2, heterog. (B)	order	0 (by arg.)	1.000	0.000	0.429	0.493
no signal (rqa)	65 × 2, heterog. (B)	jitter 160 px	+0.001 (0.012)	1.000	0.000	0.498	0.493
no signal (rqa)	65 × 2, heterog. (B)	jitter 40 px	+0.005 (0.011)	1.000	0.000	0.451	0.493
no signal (rqa)	65, 16:49 (B)	order	0 (by arg.)	0.925	0.075	0.502	0.474
no signal (rqa)	65, 16:49 (B)	jitter 160 px	+0.005 (0.012)	1.000	0.000	0.605	0.474
no signal (rqa)	65, 16:49 (B)	jitter 40 px	+0.005 (0.012)	1.000	0.000	0.573	0.474
no signal (rqa)	65, unequal (B)	order	0 (by arg.)	1.000	0.000	0.389	0.508
no signal (rqa)	65, unequal (B)	jitter 160 px	+0.014 (0.008)	1.000	0.000	0.455	0.508

Design	Size	Contrast	Target θ (SE)	Coverage	Excl. 0	Width	Obs. AUC
no signal (rqa)	65, unequal (B)	jitter 40 px	+0.016 (0.008)	1.000	0.000	0.446	0.508
no signal (rqa)	60 × 4, heterog. (W)	order	0 (by arg.)	1.000	0.000	0.377	0.508
no signal (rqa)	60 × 4, heterog. (W)	jitter 160 px	−0.009 (0.006)	1.000	0.000	0.375	0.508
no signal (rqa)	60 × 4, heterog. (W)	jitter 40 px	−0.007 (0.006)	1.000	0.000	0.362	0.508
no signal (rqa)	60, 1:3 (W)	order	0 (by arg.)	1.000	0.000	0.393	0.510
no signal (rqa)	60, 1:3 (W)	jitter 160 px	+0.010 (0.009)	1.000	0.000	0.445	0.510
no signal (rqa)	60, 1:3 (W)	jitter 40 px	+0.007 (0.009)	1.000	0.000	0.432	0.510
no signal (rqa)	60, unequal (W)	order	0 (by arg.)	1.000	0.000	0.295	0.490
no signal (rqa)	60, unequal (W)	jitter 160 px	+0.012 (0.006)	1.000	0.000	0.349	0.490
no signal (rqa)	60, unequal (W)	jitter 40 px	+0.008 (0.006)	1.000	0.000	0.334	0.490
weak order contrast (directed)	65 × 2 (B)	order	+0.118 (0.008)	1.000	0.075	0.424	0.632
weak order contrast (directed)	65 × 2 (B)	jitter 40 px	+0.015 (0.007)	1.000	0.000	0.425	0.632
weak order contrast (directed)	65 × 2 (B)	rotation	+0.000 (0.006)	1.000	0.000	0.373	0.632
weak order contrast (directed)	60 × 4 (W)	order	+0.128 (0.007)	0.975	0.350	0.307	0.639
weak order contrast (directed)	60 × 4 (W)	jitter 40 px	+0.015 (0.005)	1.000	0.000	0.298	0.639
weak order contrast (directed)	60 × 4 (W)	rotation	−0.004 (0.005)	1.000	0.000	0.265	0.639
weak spread contrast (rqa)	65 × 2, heterog. (B)	order	0 (by arg.)	1.000	0.000	0.282	0.671
weak spread contrast (rqa)	65 × 2, heterog. (B)	jitter 40 px	+0.127 (0.007)	1.000	0.125	0.412	0.671
weak spread contrast (rqa)	65, 16:49 (B)	order	0 (by arg.)	1.000	0.000	0.172	0.871
weak spread contrast (rqa)	65, 16:49 (B)	jitter 40 px	+0.315 (0.005)	1.000	0.975	0.394	0.871
weak spread contrast (rqa)	65, unequal (B)	order	0 (by arg.)	1.000	0.000	0.106	0.878
weak spread contrast (rqa)	65, unequal (B)	jitter 40 px	+0.310 (0.004)	1.000	1.000	0.302	0.878
weak spread contrast (rqa)	65 × 2 (B)	order	0 (by arg.)	1.000	0.000	0.133	0.882
weak spread contrast (rqa)	65 × 2 (B)	jitter 40 px	+0.306 (0.004)	1.000	1.000	0.334	0.882
weak spread contrast (rqa)	60 × 4, heterog. (W)	order	0 (by arg.)	1.000	0.000	0.152	0.693
weak spread contrast (rqa)	60 × 4, heterog. (W)	jitter 40 px	+0.143 (0.003)	1.000	0.600	0.250	0.693
weak spread contrast (rqa)	60, 1:3 (W)	order	0 (by arg.)	1.000	0.000	0.093	0.872
weak spread contrast (rqa)	60, 1:3 (W)	jitter 40 px	+0.302 (0.004)	0.975	1.000	0.290	0.872
weak spread contrast (rqa)	60, unequal (W)	order	0 (by arg.)	1.000	0.000	0.059	0.881
weak spread contrast (rqa)	60, unequal (W)	jitter 40 px	+0.300 (0.003)	1.000	1.000	0.209	0.881
weak spread contrast (rqa)	60 × 4 (W)	order	0 (by arg.)	1.000	0.000	0.070	0.874
weak spread contrast (rqa)	60 × 4 (W)	jitter 40 px	+0.302 (0.003)	1.000	1.000	0.247	0.874
signal elsewhere, no offset (rqa)	65 × 2, heterog. (B)	order	0 (by arg.)	1.000	0.000	0.299	0.637
signal elsewhere, no offset (rqa)	65, 16:49 (B)	order	0 (by arg.)	1.000	0.000	0.213	0.824
signal elsewhere, no offset (rqa)	65, unequal (B)	order	0 (by arg.)	1.000	0.000	0.126	0.836
signal elsewhere, no offset (rqa)	60 × 4, heterog. (W)	order	0 (by arg.)	1.000	0.000	0.177	0.660
signal elsewhere, no offset (rqa)	60, 1:3 (W)	order	0 (by arg.)	1.000	0.000	0.113	0.833
signal elsewhere, no offset (rqa)	60, unequal (W)	order	0 (by arg.)	1.000	0.000	0.072	0.841

Table S17: Bootstrap standard error against the empirical spread of the estimate in the calibration campaign ($R = 6$, $B = 60$; 40 datasets per design and size). SE ratio: mean bootstrap SE divided by the SD of the estimate across datasets, per contrast; median and range over the contrasts of the cell. Coverage of the target by normal intervals at nominal 80%, 90% and 95%.

Design	Size	Contrasts	SE ratio, median (range)	Coverage at 80%	at 90%	at 95%
no signal	65 × 2 (B)	8	1.66 (1.35–2.04)	0.959	0.988	0.997
no signal	95 × 20 (B)	8	1.48 (1.23–2.03)	0.922	0.978	0.988
no signal	112 × 2 (W)	8	1.80 (1.66–2.02)	0.963	0.994	1.000
no signal	60 × 4 (W)	8	1.80 (1.48–2.22)	0.966	0.987	1.000
signal elsewhere	65 × 2 (B)	5	2.40 (1.94–3.10)	0.985	0.990	1.000
signal elsewhere	95 × 20 (B)	5	1.46 (1.28–1.65)	0.925	0.985	1.000
signal elsewhere	112 × 2 (W)	5	2.35 (2.04–2.82)	0.990	1.000	1.000
signal elsewhere	60 × 4 (W)	5	2.65 (1.76–3.37)	0.995	1.000	1.000
signal elsewhere, no offset	65 × 2 (B)	3	2.40 (1.87–2.58)	1.000	1.000	1.000
signal elsewhere, no offset	95 × 20 (B)	3	1.29 (1.28–1.66)	0.925	0.967	0.983
signal elsewhere, no offset	112 × 2 (W)	3	1.97 (1.57–2.82)	0.975	1.000	1.000
signal elsewhere, no offset	60 × 4 (W)	3	2.19 (1.67–3.37)	0.975	1.000	1.000

Table S18: Corrected refit bootstrap (Equation 1) against the direct refit bootstrap, which averages all $R = 6$ transformed replicates inside each of the $B = 60$ draws, on the same 200 datasets per cell. Coverage of the target, mean width, and the share of intervals excluding zero.

				Coverage		Width		Excl. 0	
Design	Size	Descr.	Contrast	direct	corr.	direct	corr.	direct	corr.
no signal	65 × 2 (B)	rqa	order	0.985	0.990	0.367	0.434	0.015	0.010
no signal	65 × 2 (B)	rqa	jitter 160 px	0.995	1.000	0.419	0.505	0.005	0.000
no signal	65 × 2 (B)	rqa	jitter 40 px	0.985	0.990	0.417	0.498	0.015	0.010
no signal	60 × 4 (W)	classic	order	0.995	0.985	0.241	0.286	0.005	0.015
no signal	60 × 4 (W)	rqa	order	0.990	0.995	0.272	0.324	0.010	0.005
no signal	60 × 4 (W)	rqa	jitter 160 px	0.995	1.000	0.313	0.379	0.005	0.005
no signal	60 × 4 (W)	rqa	jitter 40 px	0.990	1.000	0.314	0.376	0.010	0.000
signal elsewhere	65 × 2 (B)	rqa	order	1.000	1.000	0.125	0.161	0.000	0.000
signal elsewhere	65 × 2 (B)	rqa	jitter 40 px	1.000	1.000	0.219	0.345	1.000	0.955
signal elsewhere	60 × 4 (W)	rqa	order	1.000	1.000	0.070	0.089	0.000	0.000
signal elsewhere	60 × 4 (W)	rqa	jitter 40 px	0.995	1.000	0.147	0.238	1.000	1.000
signal elsewhere, no offset	60 × 4 (W)	rqa	order	1.000	1.000	0.070	0.089	0.000	0.000

Table S20: Controls for the EMIP program and correctness contrasts (within-fold AUC). Program contrast: AUC on real gaze and on synthetic gaze, in which every trial is replaced by the same number of fixations drawn uniformly over the bounding box of its program's code tokens. Correctness, both programs: AUC over all correct–incorrect pairs of a test fold, the mean of its permutation distribution when labels are permuted within programs, and the AUC over pairs from the same program and test fold with a refit participant-bootstrap interval (200 draws) and a one-sided permutation p-value (1000 permutations within programs). Descriptors with no spatial features are not affected by the synthetic control and are marked with a dash.

	Program contrast		Correctness, both programs		
Descriptor	Real gaze	Synthetic gaze	All pairs	Perm. mean	Same-program pairs [95% interval]
Count	–	–	0.495	0.515	0.517 [0.317, 0.646] ($p = 0.455$)
Dispersion	0.957	1.000	0.716	0.710	0.490 [0.398, 0.581] ($p = 0.594$)
Classic	–	–	0.501	0.505	0.523 [0.319, 0.625] ($p = 0.394$)
Dur. trend	–	–	0.511	0.502	0.538 [0.373, 0.640] ($p = 0.262$)
RQA $0.5u$	0.580	0.604	0.571	0.517	0.527 [0.403, 0.645] ($p = 0.304$)
RQA $1u$	0.547	0.648	0.525	0.502	0.509 [0.384, 0.636] ($p = 0.462$)
Directed	0.655	0.554	0.580	0.540	0.562 [0.410, 0.629] ($p = 0.130$)
FAG lattice	0.658	0.859	0.664	0.566	0.577 [0.484, 0.685] ($p = 0.042$)
FAG AOI	0.778	0.879	0.635	0.619	0.559 [0.441, 0.666] ($p = 0.087$)
FAG AOI pad.	0.729	0.789	0.571	0.591	0.494 [0.345, 0.603] ($p = 0.575$)

Table S21: EMIP expertise: AUC of each of the ten test folds and of the corresponding training set, for three descriptors (linear SVM; class-balanced folds).

Fold	Classic		Directed		FAG AOI	
	test	train	test	train	test	train
1	0.686	0.532	0.400	0.593	0.229	0.588
2	0.167	0.582	0.033	0.616	0.300	0.533
3	0.143	0.595	0.381	0.589	0.095	0.598
4	0.929	0.497	0.964	0.485	0.107	0.523
5	0.429	0.551	0.257	0.594	0.200	0.595
6	0.000	0.605	0.133	0.608	0.467	0.532
7	0.300	0.562	0.500	0.576	0.100	0.549
8	0.733	0.528	0.367	0.539	0.533	0.531
9	0.214	0.533	0.214	0.538	0.286	0.542
10	0.844	0.528	0.438	0.569	0.344	0.567
Mean	0.444	0.551	0.369	0.571	0.266	0.556

Table S22: Share of dwell time on the AOI triangulation, negative / positive class, for every EMIP problem and transformation (mean over trials of the mean over $R = 6$ replicates). Fixations off the triangulation are dropped by the AOI graph. On the lattice graph and on the padded AOI graph the share was at least 0.998 in every cell.

Transformation	`em_program`	`em_halves`	`em_expertise`	`em_correct`	`em_correctrect`	`em_correctveh`
none (observed)	0.974 / 0.975	0.960 / 0.990	0.977 / 0.973	0.973 / 0.974	0.972 / 0.972	0.978 / 0.975
order	0.974 / 0.975	0.960 / 0.990	0.977 / 0.973	0.973 / 0.974	0.972 / 0.972	0.978 / 0.975
duration coupling	0.975 / 0.977	0.962 / 0.990	0.978 / 0.975	0.974 / 0.975	0.973 / 0.972	0.978 / 0.977
duration marginal	0.975 / 0.977	0.962 / 0.990	0.978 / 0.975	0.974 / 0.975	0.973 / 0.972	0.979 / 0.977
rotation	0.901 / 0.767	0.788 / 0.888	0.838 / 0.814	0.864 / 0.816	0.897 / 0.882	0.742 / 0.785
jitter $0.25u$	0.970 / 0.970	0.956 / 0.983	0.971 / 0.968	0.968 / 0.970	0.968 / 0.968	0.971 / 0.970
jitter $0.5u$	0.959 / 0.957	0.945 / 0.971	0.956 / 0.958	0.955 / 0.958	0.955 / 0.960	0.957 / 0.958
jitter $1u$	0.927 / 0.922	0.911 / 0.937	0.920 / 0.923	0.921 / 0.927	0.920 / 0.931	0.917 / 0.924
jitter $2u$	0.852 / 0.843	0.829 / 0.867	0.841 / 0.844	0.843 / 0.851	0.845 / 0.861	0.831 / 0.850

Table S23: AOI-graph sensitivities on the two halves of a program reading under the default transformations and under support-matched transformations (redrawn until the trial keeps at least its observed dwell share minus 0.02; in brackets, the share of draws for which this succeeded within 200 tries), and the sensitivities of a descriptor made only of the fixation and dwell shares on the support. In the first column: observed AUC of the AOI graph and of the support-only descriptor. Bold: interval excludes zero.

Problem	Contrast	AOI graph	AOI graph, support-matched	Support-only descriptor
`em_halves` (AUC 0.682; 0.773)	rotation	**+0.098 [+0.031, +0.166]**	+0.032 [−0.059, +0.124] (99%)	+0.054 [−0.001, +0.110]
`em_halves` (AUC 0.682; 0.773)	jitter $0.25u$	+0.036 [−0.029, +0.102]	+0.019 [−0.048, +0.086] (100%)	**+0.048 [+0.018, +0.078]**
`em_halves` (AUC 0.682; 0.773)	jitter $0.5u$	+0.023 [−0.025, +0.071]	+0.035 [−0.029, +0.099] (100%)	**+0.112 [+0.065, +0.159]**
`em_halves` (AUC 0.682; 0.773)	jitter $1u$	+0.045 [−0.032, +0.123]	+0.057 [−0.027, +0.141] (89%)	**+0.119 [+0.059, +0.178]**
`em_halves` (AUC 0.682; 0.773)	jitter $2u$	**+0.113 [+0.014, +0.212]**	**+0.121 [+0.035, +0.207]** (31%)	**+0.123 [+0.054, +0.192]**

Table S24: Duration-marginal transformation of the classic descriptor with the leave-one-participant-out donor pool of the main analysis and with training-fold pools (each fold's trials draw only from that fold's training participants; mean and SD over $R = 6$ replicates; within-fold AUC, class-balanced folds).

Problem	AUC obs.	AUC, LOO pool	AUC, training-fold pool (SD)	C LOO	C training-fold
gb_tex_vd1	0.969	0.935	0.935 (0.000)	+0.035	+0.034
gb_vd1_blg	0.886	0.852	0.852 (0.001)	+0.033	+0.034
gb_vd1_vd2	0.640	0.516	0.523 (0.009)	+0.124	+0.117
gb_tex_s1s2	0.539	0.508	0.503 (0.020)	+0.031	+0.036
gb_vd1_halves	0.564	0.556	0.553 (0.004)	+0.008	+0.011
sb_native	0.689	0.672	0.672 (0.002)	+0.017	+0.017
sb_compr	0.587	0.561	0.562 (0.005)	+0.026	+0.024
em_program	0.539	0.507	0.532 (0.010)	+0.032	+0.007
em_expertise	0.438	0.461	0.474 (0.038)	−0.023	−0.037
em_correct	0.501	0.447	0.454 (0.023)	+0.054	+0.047
em_correctrect	0.523	0.474	0.411 (0.050)	+0.048	+0.112
em_correctveh	0.500	0.475	0.454 (0.070)	+0.025	+0.046
em_halves	0.776	0.772	0.772 (0.005)	+0.004	+0.004

Table S25: Three permutation distributions for negative sensitivities (within-fold AUC, class-balanced folds; 1000 permutations under the schemes of Table 2 of the article; $R = 6$ transformed replicates). The observed AUC, the transformed AUC and their contrast are each compared with their own permutation distribution; for the contrast the lower-tail probability is given. The first four rows are the order cells of the classic descriptor discussed in Section 5.4 of the article; the last is the unexplained cell of Section 5.11.

Problem	Descriptor	Transformation	Observed AUC	p	Transformed AUC	p	Contrast	p (lower tail)
em_program	Classic	order	0.539	0.095	0.788	0.001	−0.249	0.001
gb_vd1_halves	Classic	order	0.564	0.001	0.653	0.001	−0.090	0.001
gb_tex_vd1	Classic	order	0.969	0.001	0.991	0.001	−0.022	0.142
gb_vd1_blg	Classic	order	0.886	0.001	1.000	0.001	−0.114	0.001
em_halves	RQA $0.5u$	jitter $1u$	0.613	0.001	0.692	0.001	−0.080	0.027

Table S26: Calibration at the largest analysed sample size: 322 participants × 4 trials with the label varying within participants ($R = 6$, $B = 60$; 40 datasets per row; targets from 60 further datasets with $R = 20$, or zero by exchangeability for the order contrast). Coverage of the target by the 95% interval; exclusions of zero where the target is zero by argument; mean width; mean bootstrap SE over the empirical SD of the estimate; coverage of the nominal 80% interval.

Design	Descriptor	Contrast	Coverage	Excl. 0 at zero target	Width	SE ratio	Coverage at 80%
no signal	classic	order	40/40	0/40	0.111	1.65	0.975
no signal	directed	order	38/40	2/40	0.139	1.39	0.950
no signal	directed	jitter 160 px	39/40	–	0.148	1.26	0.900
no signal	directed	jitter 40 px	39/40	–	0.150	1.39	0.975
no signal	directed	rotation	40/40	–	0.144	1.42	0.900
no signal	rqa	order	40/40	0/40	0.133	1.63	0.950
no signal	rqa	jitter 160 px	40/40	–	0.159	1.70	1.000
no signal	rqa	jitter 40 px	40/40	–	0.150	1.62	0.925
signal elsewhere	directed	order	40/40	0/40	0.024	1.38	0.900
signal elsewhere	directed	jitter 40 px	40/40	–	0.075	1.97	1.000
signal elsewhere	directed	rotation	40/40	–	0.045	1.44	0.975
signal elsewhere	rqa	order	40/40	0/40	0.017	1.57	1.000
signal elsewhere	rqa	jitter 40 px	40/40	–	0.070	1.81	0.975

Table S27: Gradient-boosted trees on the cells of Table 7 of the article at the screening settings and at the long settings (seed 42). Bold: interval excludes zero.

Problem	Descriptor	Transformation	Trees, $R = 6$, $B = 60$	Trees, $R = 20$, $B = 300$
gb_tex_vd1	Dur. trend	duration marginal	**+0.371 [+0.318, +0.424]**	**+0.378 [+0.327, +0.428]**
gb_vd1_blg	Classic	order	**−0.118 [−0.148, −0.088]**	**−0.118 [−0.146, −0.090]**
gb_vd1_blg	Dur. trend	duration marginal	**+0.226 [+0.161, +0.291]**	**+0.231 [+0.168, +0.293]**
gb_vd1_vd2	Classic	duration marginal	**+0.130 [+0.073, +0.187]**	**+0.127 [+0.074, +0.180]**
gb_vd1_vd2	Dispersion	jitter $2u$	**+0.102 [+0.038, +0.166]**	**+0.100 [+0.036, +0.164]**
gb_vd1_vd2	Dur. trend	duration coupling	**+0.176 [+0.133, +0.219]**	**+0.172 [+0.128, +0.215]**
gb_vd1_vd2	Dur. trend	duration marginal	**+0.272 [+0.216, +0.329]**	**+0.274 [+0.226, +0.322]**
gb_vd1_vd2	Dur. trend	order	**+0.178 [+0.133, +0.224]**	**+0.175 [+0.132, +0.218]**
gb_vd1_vd2	RQA $0.5u$	jitter $2u$	**+0.106 [+0.041, +0.170]**	**+0.103 [+0.037, +0.168]**
gb_vd1_vd2	RQA $1u$	jitter $2u$	**+0.125 [+0.054, +0.195]**	**+0.123 [+0.067, +0.178]**
gb_vd1_halves	Directed	jitter $2u$	**+0.192 [+0.140, +0.243]**	**+0.199 [+0.145, +0.252]**
gb_vd1_halves	Dispersion	jitter $2u$	**+0.123 [+0.097, +0.150]**	**+0.124 [+0.096, +0.153]**
gb_vd1_halves	Dispersion	rotation	**+0.177 [+0.139, +0.215]**	**+0.167 [+0.133, +0.202]**
gb_vd1_halves	RQA $0.5u$	jitter $2u$	**+0.157 [+0.101, +0.214]**	**+0.165 [+0.108, +0.222]**
gb_vd1_halves	RQA $1u$	jitter $2u$	**+0.212 [+0.172, +0.253]**	**+0.219 [+0.168, +0.270]**
sb_native	RQA $0.5u$	jitter $2u$	−0.002 [−0.084, +0.081]	+0.008 [−0.074, +0.091]
sb_native	RQA $1u$	jitter $2u$	**+0.100 [+0.009, +0.190]**	**+0.111 [+0.024, +0.199]**
sb_native	RQA $1u$	jitter $1u$	+0.071 [−0.003, +0.145]	**+0.082 [+0.012, +0.151]**
em_program	Dispersion	rotation	**+0.166 [+0.088, +0.244]**	**+0.164 [+0.074, +0.255]**
em_halves	Classic	order	**+0.128 [+0.033, +0.222]**	**+0.148 [+0.062, +0.233]**
em_halves	RQA $0.5u$	jitter $1u$	−0.089 [−0.189, +0.011]	−0.087 [−0.192, +0.018]

Methods appendix

A1. Feature definitions

Let a trial have fixations (x_i, y_i, d_i), $i = 1, \ldots, N$, in sequence order, with d_i the duration.

Count. N.

Dispersion. The population SDs of x and of y, and the mean pairwise distance $\binom{N}{2}^{-1} \sum_{i<j} \|p_i - p_j\|$, $p_i = (x_i, y_i)$.

Classic. Mean duration $N^{-1} \sum_i d_i$ and mean saccade amplitude $(N-1)^{-1} \sum_{i<N} \|p_{i+1} - p_i\|$.

Duration trend. The Pearson correlation of d_i with i, and the mean duration; trials with fewer than five fixations are excluded.

Recurrence (Anderson et al., 2013). Fixations $i < j$ recur if $\|p_i - p_j\| < r$; n_r is the number of recurrent pairs, and the recurrence plot is the upper triangle of the $N \times N$ recurrence matrix. Recurrence rate: $100 \cdot 2n_r/(N(N-1))$. Determinism: 100 times the share of recurrent pairs that lie on diagonal lines of length at least 2. Laminarity: 100 times the share that lie on vertical lines of length at least 2. Centre of recurrence mass: $100 \cdot \overline{(j-i)}/(N-1)$, the mean over recurrent pairs. A trial without recurrent pairs has all four equal to zero; trials with fewer than five fixations are excluded. Radii: $r = 0.5u$ and $1u$.

Directed transition graph. Each fixation is assigned to the tile $(\lfloor x/u \rfloor, \lfloor y/u \rfloor)$; consecutive repeats are collapsed to a sequence s over the visited tiles V; A_{ab} counts immediate transitions $a \to b$, and $E = \{(a, b) : a \neq b, A_{ab} > 0\}$. Reciprocity: the share of edges in E whose reverse is in E. Revisit tendency: $1 - |V|/|s|$. Transition entropy: $-\sum_{ab} P_{ab} \log_2 P_{ab}$ with $P = A/\sum A$. Degree asymmetry: the mean over tiles of $|\mathrm{out}_a - \mathrm{in}_a|/(\mathrm{out}_a + \mathrm{in}_a)$, with out- and in-strengths the row and column sums of A.

Fixation Adjacency Graph. Anchors are triangulated (Delaunay). The dwell of a triangle is the summed duration of the fixations inside it; fixations in no triangle are dropped. Vertices are the triangles with positive dwell; two are adjacent if they share an anchor; the weight of an edge is the summed dwell of its two triangles. Attention centrality: the largest, over connected components, of the weight of the component's maximum spanning tree, divided by the total edge weight (set to 0.25 for a graph without edges). Betweenness: the maximum edge betweenness centrality (Brandes, 2001). Compactness: the Estrada index $\sum_k e^{\lambda_k}$ of the adjacency matrix (Estrada, 2000), averaged over components. Squareness: for each component, $\Delta_x \Delta_y / \max(\Delta_x, \Delta_y)^2$ with Δ_x, Δ_y the extents of its triangle centroids (zero for a single triangle), averaged over components. The fifth feature is the number of components. Lattice anchors are $2u$ apart and cover the disc swept by the display under rotation; AOI anchors are the token centres of the program; the padded AOI graph adds lattice anchors $2u$ apart around them, leaving out those closer than $2u$ to a token anchor.

A2. Class sizes and fixation counts

Table S28: Trials and participants per class, and fixations per trial, for every problem (after the exclusions of Section 4 of the article). Class labels are those of the released loaders.

		Trials		Participants		Fixations per trial	
Problem	Positive class	positive	negative	positive	negative	median (IQR)	range
`gb_tex_vd1`	`VD1`	644	644	322	322	179 (132–225)	37–300
`gb_vd1_blg`	`BLG`	643	644	322	322	121 (96–146)	33–392
`gb_vd1_vd2`	`VD2`	644	644	322	322	141 (121–159)	32–233
`gb_vd1_halves`	`VD1#2`	644	644	322	322	66 (57–75)	18–113
`gb_tex_s1s2`	`TEX@S2`	322	322	322	322	225 (206–243)	97–300
`sb_native`	`nonnative`	620	1426	29	66	122 (92–152)	7–398
`sb_compr`	`low`	805	558	79	66	121 (93–152)	8–398
`em_program`	`vehicle`	112	112	112	112	218 (145–291)	25–868
`em_halves`	`second`	224	224	112	112	109 (72–145)	12–434
`em_expertise`	`low`	64	51	37	28	181 (129–263)	25–634
`em_correct`	`incorrect`	147	124	120	108	212 (142–286)	25–868
`em_correctrect`	`incorrect`	41	92	41	92	193 (142–279)	35–606
`em_correctveh`	`incorrect`	106	32	106	32	230 (143–293)	25–868

A3. Software and settings

Python 3.13.16; NumPy 2.5.3; SciPy 1.18.1; pandas 3.0.5; scikit-learn 1.7.2; XGBoost 3.2.0; NetworkX 3.7; package `scanpathdecomp` 0.6.3. Linear SVM: `LinearSVC` (liblinear; squared hinge loss, L2 penalty, $C = 1$, at most 20 000 iterations, random state 0) on features standardized with training-fold means and SDs. Gradient-boosted trees: `XGBClassifier` with 200 trees of depth 3, learning rate 0.05, row subsampling 0.8, histogram tree method, logistic loss, random state 0. Folds: ten participant folds, participants dealt within class in a seeded order (seed 0 for the reported profiles; seeds 0–19 for the partition check).

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